

OPTIMIZING ENERGY CONVERSION IN STEADY-STATE NONTHERMAL QUANTUM DEVICES

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1. Introduction

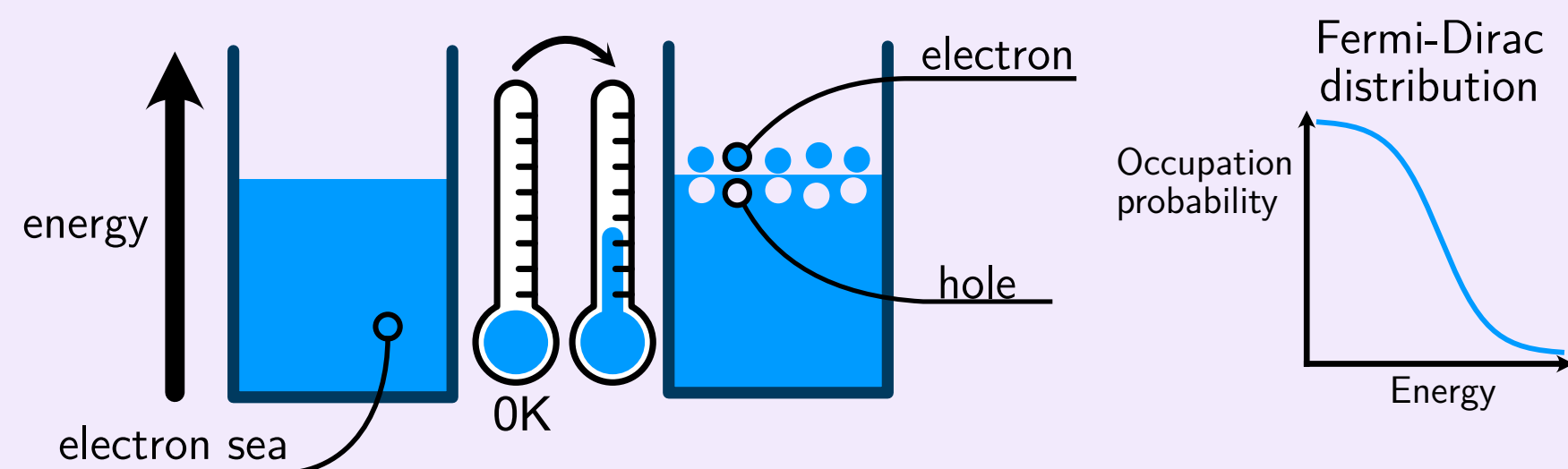
In quantum transport, one investigates particle currents through quantum devices coupled to multiple reservoirs, which are defined by their chemical potentials and temperatures. Recently, there has been a large interest in studying the energy conversion processes in these devices, making the quantum thermodynamical aspects of them relevant. For example, one could imagine using dissipated heat from regular processes as a resource to cool critical components on nanochips. However, when reaching the nanoscale, particles might no longer equilibrate with their thermal surroundings. Consequently, to investigate energy conversion processes, nonthermal distributions become highly relevant descriptors of the particles' environment.

Using Landauer scattering theory, the optimization is made by adjusting the transmission probabilities of electrons at different energies. We also address the issue of how to define an efficiency as the energy current cannot be neatly divided into heat and work by using the entropy currents [2] as output and resource.

What is the best-performing refrigerator?

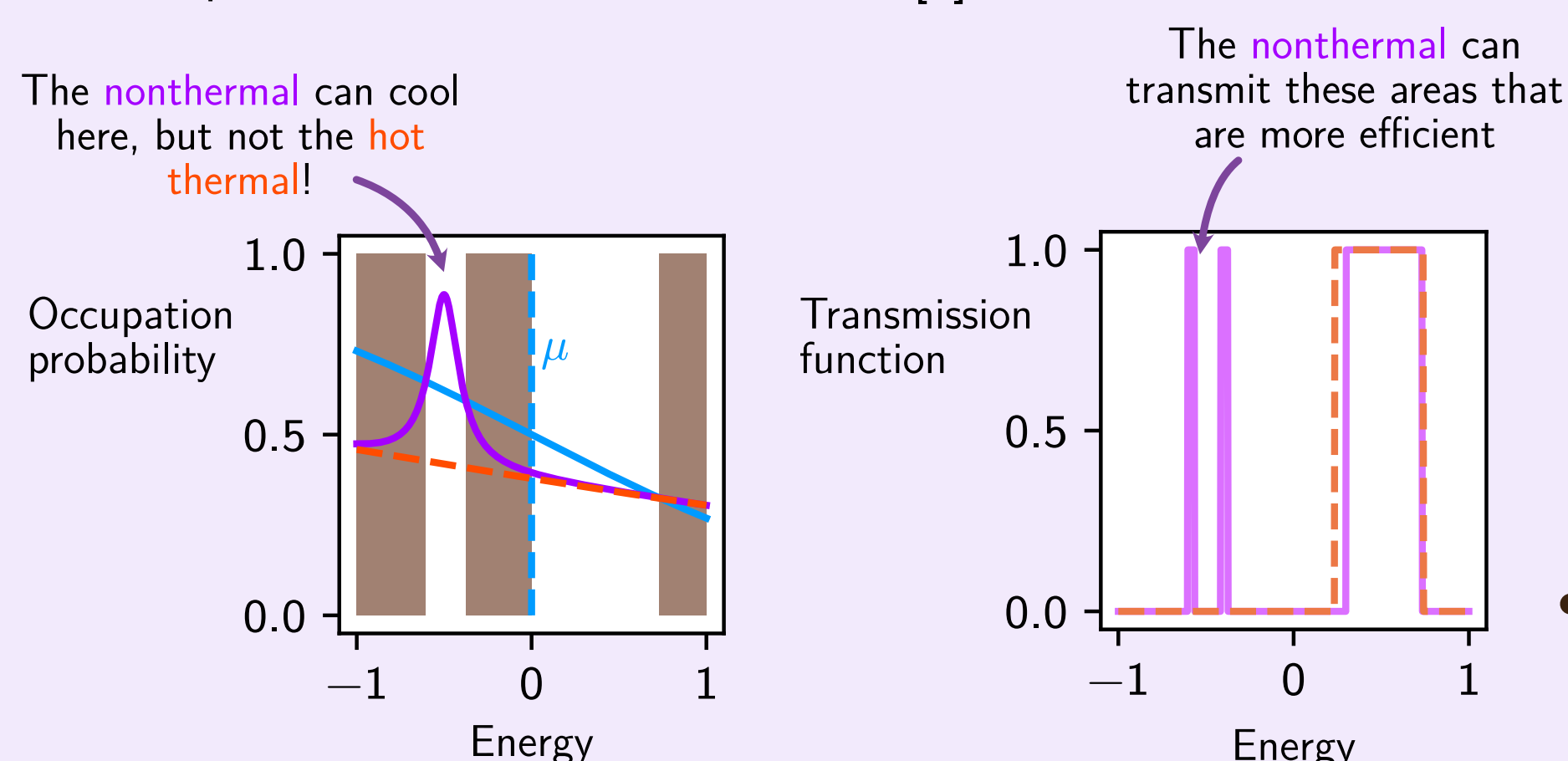
Thermal distribution

The temperature in a thermal bath depends on the amount of excited electrons above the electrochemical potential.



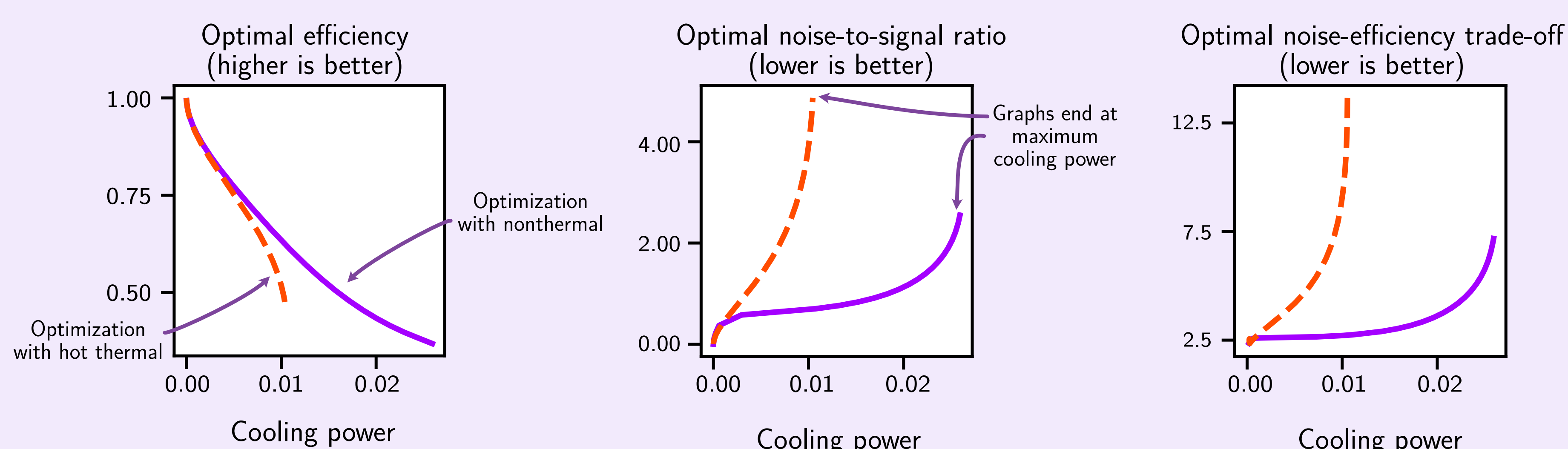
3. Example optimization

The optimization picks out boxcars that give the best efficiency for a certain cooling power. Compare with the equivalent hot thermal distribution [5].



4. Nonthermal performs better than thermal

The optimal transmission functions are found for every possible cooling power for both the nonthermal and hot thermal distributions exchanging with the cold distribution.



In all performance metrics, the nonthermal bath outperforms the thermal one, when using optimized transmission functions

- References
- [1] In preparation, E. Danielsson, H. Kirchberg, J. Splettstoesser
 - [2] M. Acciai et al., Phys. Rev. B, 109, (2024)
 - [3] R. S. Whitney, Phys. Rev. B, 91 (2015)
 - [4] A. M. Timpanaro et al., Phys. Rev. B 111, 014301 (2025)
 - [5] M. Buttiker, Phys. Rev. B 33, 3020 (1986); Phys. Rev. B 32, 1846 (1985)

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2. Analytical optimization

General current in scattering theory:

$$I^x = \int d\varepsilon x(\varepsilon) \mathcal{D}(\varepsilon) (f(\varepsilon) - g(\varepsilon))$$

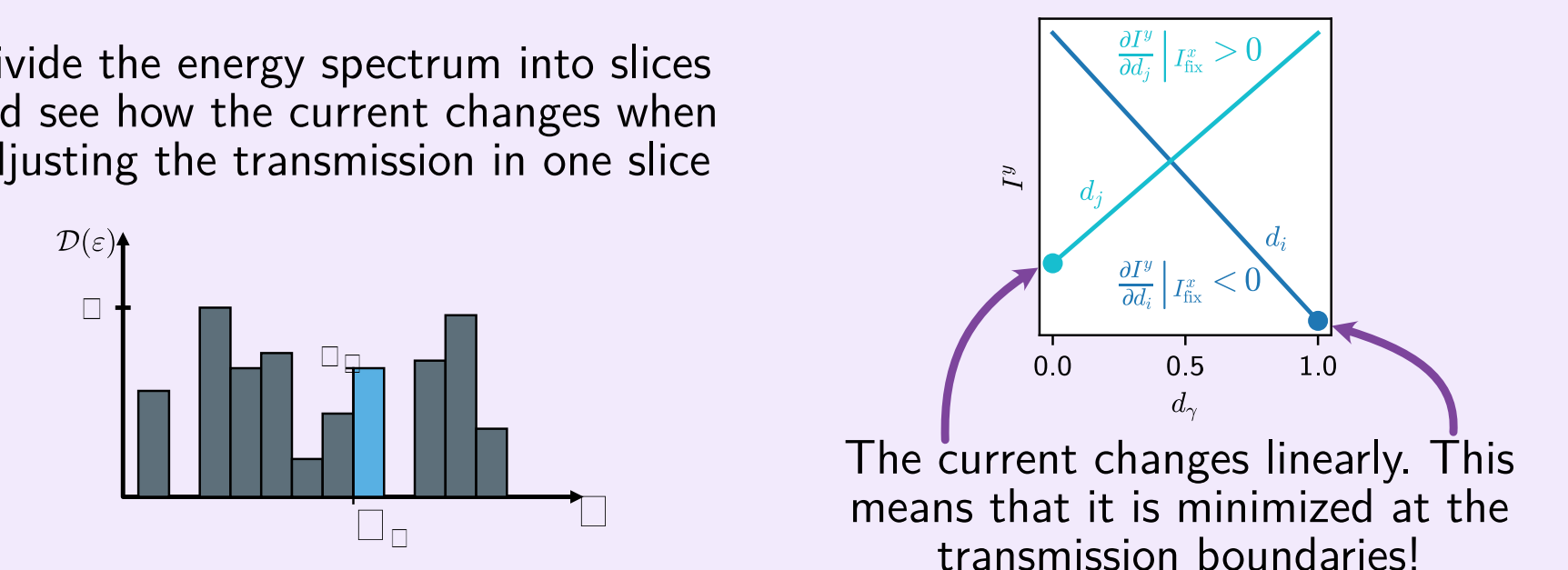
Transmission function $\in [0,1]$ (blue arrow)
Current coefficient (energy, heat, entropy...) (red arrow)
Thermal dist. to cool (blue arrow)
Nonthermal distribution (purple arrow)

Goal: Design the transmission function to get the highest efficiency for a fixed output current

$$\text{Efficiency} \rightarrow \eta = \frac{I^x}{I^y}$$

Fixed output (blue arrow)
Resource current to minimize (red arrow)

Divide the energy spectrum into slices and see how the current changes when adjusting the transmission in one slice

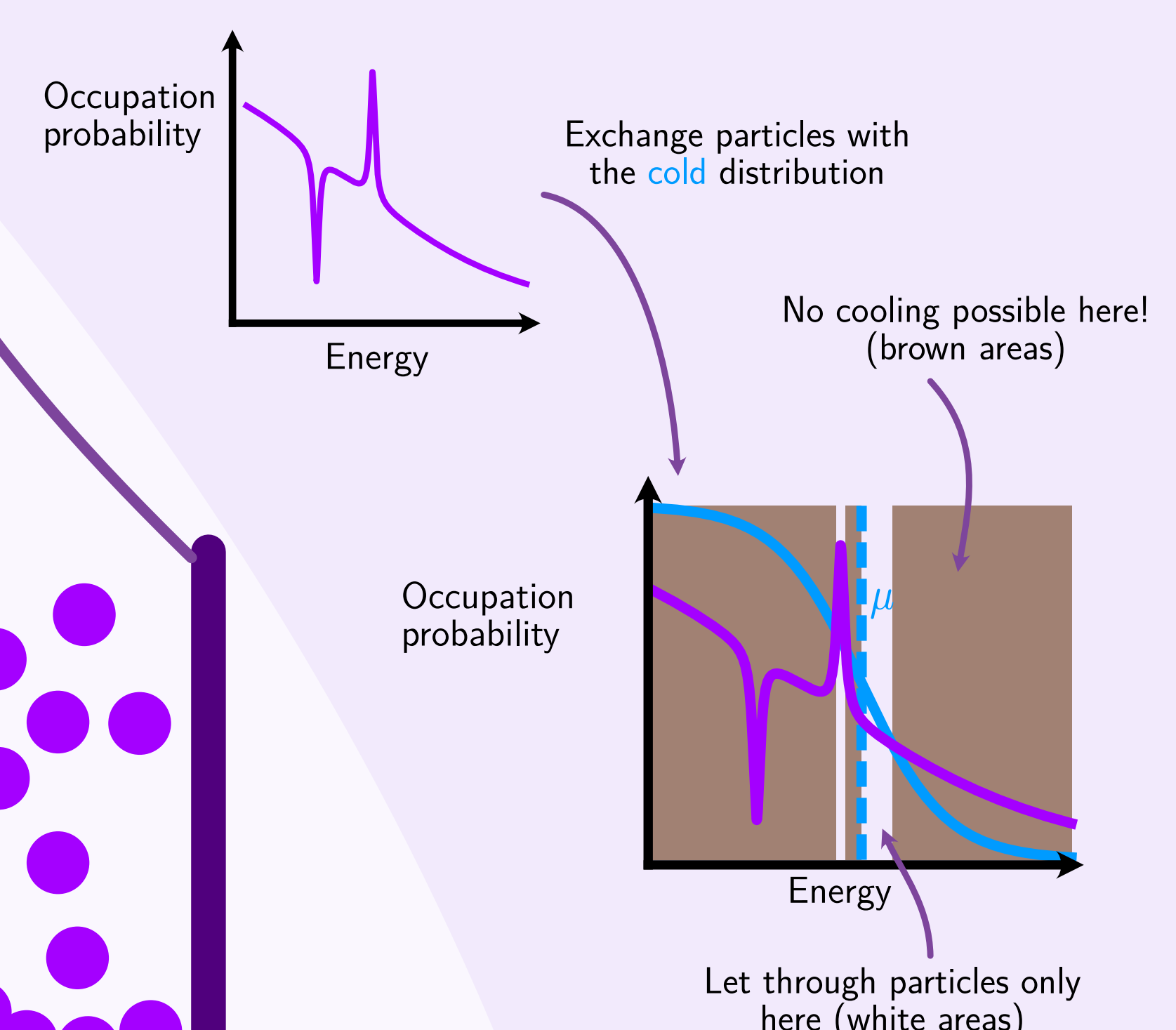


Solution: Transmit the electrons that contribute *most efficiently* to the cooling power and block the rest

The optimal transmission function is a series of boxcars

This conclusion holds also for **minimizing noise** and **optimizing precision-efficiency trade-offs**, but with different positions and widths of the boxcars.

Nonthermal distribution



6. Conclusion

Nonthermal reservoirs can be exploited for energy conversion better than thermal counterparts. This is shown by optimizing the transmission function, which is achievable for any task and goal.

Insights

- The optimal transmission function for any efficiency, noise or trade-off is a series of boxcars, determined by crossing points in the current integrands.
- These functions are different for a nonthermal distribution and its thermal counterpart, underlining the importance of device design with nonthermality in mind.
- The optimization provides bounds on device performance, informing what is fundamentally achievable for an energy converter

Future goals

- Investigating the input: How can we quantify the resourcefulness of the nonthermal reservoir
- Experimental applications: Where and when does nonthermality occur and can we exploit it in reality?